ViFaithRAG: A Vietnamese RAG Benchmark for Faithfulness on Real-World Queries

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***Abstract*—Retrieval-augmented generation (RAG) can reduce factual hallucination, yet models may still contradict retrieved evidence or attempt to answer when that evidence is insufficient. Existing benchmarks often use constructed questions or curated tasks, leaving this failure under realistic web retrieval understudied. We introduce ViFaithRAG, a Vietnamese RAG faithfulness benchmark built from 4,117 real legal and open-domain queries and the full text of their top three Google results. Each record has a human-verified YES, PARTIAL, NO, or AMBIGUOUS answerability label, supporting-document annotations, and a reviewed reference response. Independent annotation reaches 87.35% four-way agreement and a Cohen's kappa of 0.744, while 35.3% of records are not fully and unambiguously answerable. Across four baseline models, the best observed scores are 78.54% reference-based faithfulness and 79.55% abstention F1, achieved by different models; even the model with the lowest scope-violation rate answers beyond the supported scope on 22.39% of PARTIAL records. These results show that faithful RAG evaluation must measure both grounded generation and the decision to qualify, clarify, or abstain.**

***Index Terms*—retrieval-augmented generation, faithfulness, hallucination, answerability, Vietnamese benchmark**

I. Introduction

Retrieval-augmented generation (RAG) grounds large language models (LLMs) in external knowledge, but retrieval alone does not ensure faithfulness: a model may contradict the evidence, add unsupported claims, or answer when the retrieved pages are insufficient. Evaluation must therefore assess both grounded content and whether the evidence warrants an answer.

Existing benchmarks often use constructed questions or curated tasks rather than naturally occurring queries with noisy, full-page retrieval [1], [2]. The gap is pronounced in Vietnamese, where existing resources focus on fact verification or reading comprehension [3]–[5]. We introduce ViFaithRAG, built from 4,117 real legal and open-domain queries and the unfiltered full text of their top three Google Search results. Each record includes a human-verified YES, PARTIAL, NO, or AMBIGUOUS label, supporting-document annotations, and a reviewed reference response. Fig. 1 shows an example with one supporting page and two distractors.

Our main contributions are:

- **Benchmark construction.** We construct a Vietnamese RAG benchmark with LLM-assisted pre-labeling and human verification. Model pre-labels provide candidate labels, rationales, and supporting-document sets; two annotators independently label each record, with adjudication and answer review producing the final labels and reviewed reference responses.
- **Multi-dimensional faithfulness evaluation.** We propose a label-conditioned framework that separates grounded answer quality from evidence-boundary behavior.
- **Open-source LLM evaluation.** We apply the framework to popular open-source LLMs under a controlled protocol, comparing grounded generation with decisions to answer, qualify, clarify, or abstain.

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II. Related Work

Faithfulness in RAG. Hallucination can concern real-world factuality or consistency with a supplied source [6], [7]. RAG [8] addresses missing or stale knowledge, but generated claims can remain unsupported even when evidence is present [2], [9]. This motivates evaluating evidence boundaries and abstention rather than assuming that every retrieved context supports an answer.

Benchmarks. CRAG evaluates factual question answering with web and knowledge-graph retrieval, but its questions are generated from templates or written by annotators [1]. RAGTruth provides fine-grained manual hallucination annotations for RAG responses generated across several existing tasks [2]. HaluEval, FELM, FreshQA, and RGB study hallucination detection, response factuality, freshness, or retrieval noise under complementary settings [10]–[13]. ViFaithRAG instead annotates the answerability of genuine user queries paired with unfiltered full pages, allowing evaluation both before and after a model commits to an answer.

Table I shows that ViFaithRAG is the only listed resource combining Vietnamese, real user queries, unfiltered full-page contexts, and explicit graded answerability.

Vietnamese resources. ViFactCheck and ViWikiFC provide Vietnamese claim-verification data, and UIT-ViQuAD supports extractive reading comprehension [3]–[5]. Their curated claims or answer-bearing passages do not test open-ended generation from noisy search results. Among the Vietnamese resources reviewed, ViFaithRAG is the first to jointly evaluate RAG faithfulness, graded answerability, and abstention.

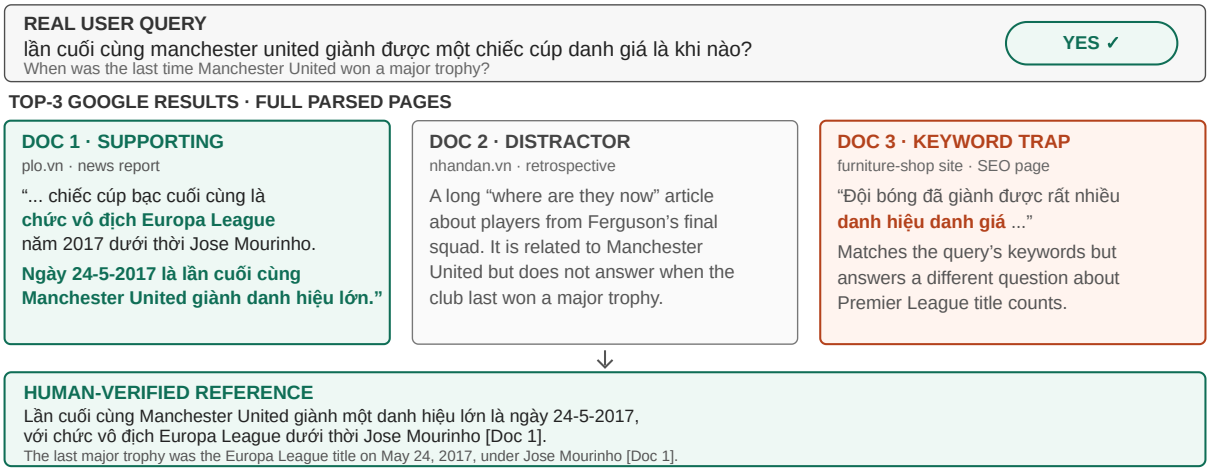


Fig. 1. A ViFaithRAG record from the retrieval snapshot. Green marks the selected supporting page and its answer-bearing evidence; gray marks a related distractor; coral marks an off-topic page that matches the query's keywords. The reviewed reference response cites only Doc 1.

TABLE I
COMPARISON ALONG THE FOUR DIMENSIONS THAT DISTINGUISH VIFAITHRAG. GRADED ANSWERABILITY DENOTES INPUT LABELS THAT SEPARATE FULLY, PARTIALLY, NOT, AND AMBIGUOUSLY ANSWERABLE QUERIES.

Benchmark	Language	Query source	Context type	Graded answerability
CRAG	English	Templates; annotator-written	Brave web; mock KG	No
RAGTruth	English	Existing task datasets	Task-specific contexts	No
HaluEval	English	Existing/synthetic data	Task-dependent	No
ViFactCheck	Vietnamese	Human-annotated claims	News articles	No
ViFaithRAG (ours)	**Vietnamese**	**Real user queries**	**Three full pages, unfiltered**	**Yes (four-way)**

III. BENCHMARK CONSTRUCTION

ViFaithRAG models the query and retrieval conditions of a deployed Vietnamese RAG system and adds evidence-sensitive human supervision. Fig. 2 summarizes the distinction from constructed-question benchmark design.

A. Queries and Retrieval

We collected approximately 30,000 naturally occurring Vietnamese queries from public Q&A pages, forums, and question portals. Rule-based filtering removed non-Vietnamese queries, queries with fewer than 4 words, queries containing personally identifiable information, and lexical near-duplicates with at least 80% word overlap. We then encoded the queries with BGE-M3 [14] and removed semantic near-duplicates above 0.85 cosine similarity. The final set contains 4,117 queries: 1,171 legal and 2,946 open-domain.

For every query, we issued a single Google Search request through the Serper API under a fixed configuration: country `vn`, interface language `vi`, Google query autocorrection enabled, results restricted to `.vn` domains, and three organic results requested. We parsed every returned page while preserving body text, headings, and tables. Because the request returns exactly the three pages we retain, the retrieval snapshot involves no result selection: beyond the query-time domain restriction, we applied no relevance filtering, passage selection, or manual page cleaning. The resulting 12,351 documents retain off-topic pages, incomplete evidence, conflicting sources, and residual boilerplate.

B. Answerability Taxonomy

Each query–context pair receives one mutually exclusive label. **YES** means the documents support every substantive part of the query; **PARTIAL** means they support at least one part but leave another unsupported; **NO** means they support no substantive answer; and **AMBIGUOUS** means a missing referent or materially different interpretations prevent a determinate answer. Time-sensitive claims count as supported only when the evidence supplies a relevant date or effective period.

C. Human Verification and References

GPT-5.5 and Gemini 3.1 Pro first produced model pre-labels, rationales, and supporting-document sets. After a 300-record calibration pilot, five trained annotators processed the full collection. *Annotator qualifications.* Annotators were recruited from a major technology company and had professional experience labeling LLM data, particularly for question-answering tasks involving legal and open-domain content. Every record received two independent human labels and supporting-document selections; annotators could consult the model pre-labels and rationales but could not see each other's decisions. A Phase 1 adjudicator resolved all 521 label disagreements without seeing the model outputs. Before adjudication, the annotators agreed

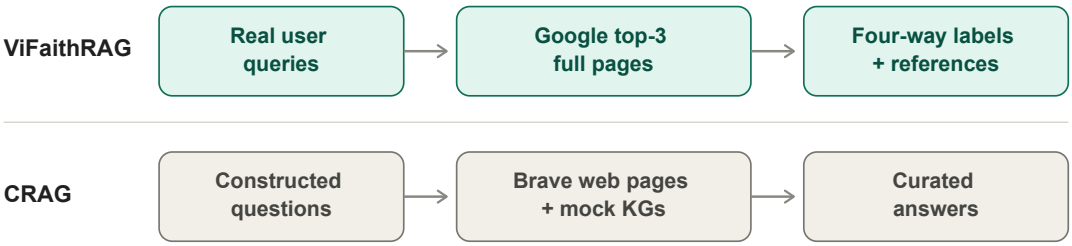


Fig. 2. Construction contrast: ViFaithRAG uses real queries and unfiltered Google top-three full pages, whereas CRAG [1] uses constructed questions, Brave Search pages, and mock knowledge graphs.

on 87.35% of four-way decisions ($\kappa = 0.744$) and 93.95% after collapsing YES/PARTIAL versus NO/AMBIGUOUS ($\kappa = 0.839$).

For YES and PARTIAL records, Gemini 3.1 Pro drafted a cited response from human-selected evidence. A human reviewer checked support, coverage, scope, consistency, and language, then accepted or edited the draft. NO and AMBIGUOUS records received reviewed abstention or clarification responses. Potential label errors discovered during answer review went to a second adjudicator. Reviewers accepted approximately 92% of model drafts; during Phase 2, the second adjudicator corrected 54 labels (1.31%). The release retains raw and final labels, supporting-document selections, reviewed reference responses, provenance, the three retrieved pages, and model pre-labels.

IV. Benchmark Characteristics

A. Answerability and Retrieval Noise

More than one third of the benchmark is not fully and unambiguously answerable. Table II reports 1,454 such records (35.3%), with similar rates in the legal and open-domain subsets. Retrieval insufficiency is therefore not confined to specialized legal questions.

The retrieved contexts combine distributed evidence with irrelevant or merely topical material. We call a page a *distractor* when annotators do not select it as direct support. Among the 2,998 YES/PARTIAL records, 82.32% draw support from multiple pages, yet 49.17% also contain at least one distractor. Queries have a median length of 10 words, while their three-page contexts have a median length of 4,378 tokens and span 2,884 domains. As Fig. 3 shows, BGE-M3 query–document similarity achieves an ROC AUC of only 0.668 for predicting support status. Topical similarity alone is therefore an imperfect proxy for answer-bearing evidence.

B. Model Pre-Labels versus Human Judgment

Model pre-labels are useful annotation aids, but they do not reliably identify evidence boundaries. Gemini 3.1 Pro agrees with final human labels on 84.2% of valid records ($\kappa = 0.680$), compared with 76.5% for GPT-5.5 ($\kappa = 0.507$); among human-labeled NO records, Gemini 3.1 Pro matches 69.8% whereas GPT-5.5 matches only 34.9%, assigning a more answerable label to most human-NO cases. Because

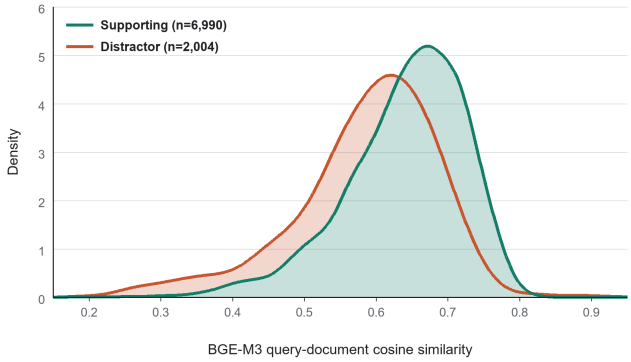


Fig. 3. Kernel density estimates of query–document cosine similarities computed using BGE-M3 dense embeddings for supporting and distractor documents in YES and PARTIAL records. Each distribution is normalized independently, and its bandwidth is selected using Silverman's rule.

annotators could consult both sets of pre-labels, these model–human agreement rates are not independent blind-validation estimates and may partly reflect pre-label exposure.

PARTIAL remains difficult across model families. GPT-5.5 and Gemini 3.1 Pro each match fewer than 52% of PARTIAL records. Under the same answerability-labeling protocol used for construction pre-labeling, DeepSeek-V4-Flash and Qwen3.5-397B-A17B achieved 78.2% and 78.6% overall agreement with the human labels, respectively, but only 21.3% and 41.9% on PARTIAL. Pre-label visibility nevertheless does not determine the released labels: final human labels differ from both GPT-5.5 and Gemini 3.1 Pro on 431 records, including 303 on which the two models agree. These cases justify retaining independent annotation and adjudication for evidence-boundary decisions.

V. RAG Faithfulness Evaluation

RAG faithfulness evaluation should assess not only whether generated claims are supported, but also whether the model should answer given the available evidence. We therefore introduce a label-conditioned LLM-as-a-judge pipeline built on the ViFaithRAG benchmark. For each record, the human-verified answerability label determines the evaluation route, while the reviewed reference response defines the supported answer scope for YES and PARTIAL records. We denote

TABLE II
ANSWERABILITY DISTRIBUTION BY DOMAIN. PARENTHESES CONTAIN ROW PERCENTAGES; "NOT FULLY" COMBINES PARTIAL, NO, AND AMBIGUOUS.

Domain	Records	YES	PARTIAL	NO	AMBIGUOUS	Not fully
Legal	1,171	779 (66.5)	80 (6.8)	302 (25.8)	10 (0.9)	392 (33.5)
Open domain	2,946	1,884 (64.0)	255 (8.7)	769 (26.1)	38 (1.3)	1,062 (36.1)
All	**4,117**	**2,663 (64.7)**	**335 (8.1)**	**1,071 (26.0)**	**48 (1.2)**	**1,454 (35.3)**

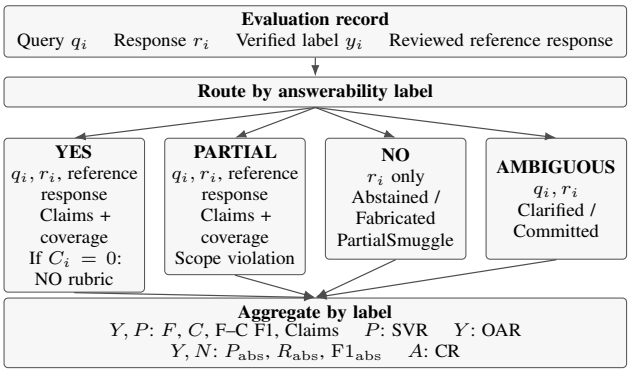


Fig. 4. Proposed label-conditioned evaluation pipeline. The verified answerability label routes each record to a specialized judge prompt, and the resulting judgments are aggregated within each label category. A YES response with $C_i = 0$ also receives the same response-only NO rubric.

the four labels by Y, P, N, and A, corresponding to YES, PARTIAL, NO, and AMBIGUOUS, respectively. For each label y, let N_y denote the number of evaluated records and N_y^b the number of records exhibiting behavior b. An evaluation is considered valid when the judge returns a parseable verdict. Fig. 4 summarizes the routing from shared inputs to label-specific outputs and aggregate metrics.

A. Label-Conditioned Judging

For YES and PARTIAL records, the judge receives the query, response, and reviewed reference response, decomposes the response into atomic claims, verifies each claim against the reference response, and measures coverage of its key points; statements about the availability of evidence rather than about the subject of the query are not extracted as claims. PARTIAL records receive an additional scope-violation verdict. NO records are classified as *Abstained*, *Fabricated*, or *PartialSmuggle*, and AMBIGUOUS records as *Clarified* or *Committed*. The judge uses only the supplied inputs and does not replace the human-verified evidence boundary with external knowledge.

A response *abstains* when it declines on the grounds that the evidence is insufficient and supplies no substantive content; Fabricated responses answer without support and PartialSmuggle responses use refusal-like language before answering anyway. The NO rubric sees only the response, and we apply that same rubric to YES responses with zero completeness, keeping only the binary outcome. Both terms of abstention precision therefore come from one label-blind classifier, and because an abstaining response covers no key point the screen omits no abstaining YES response.

B. Metrics

For record i, let $\mathcal{M}_i$ be the atomic claims and $\mathcal{G}_i$ the reference key points, with $s(m) = 1$ when claim m is supported and $c(g) = 1$ when key point g is covered:

$$F_i = \frac{\sum_{m \in \mathcal{M}_i} s(m)}{|\mathcal{M}_i|}, \qquad C_i = \frac{\sum_{g \in \mathcal{G}_i} c(g)}{|\mathcal{G}_i|}. \tag{1}$$

If $\mathcal{M}_i = \varnothing$, F_i is undefined and the record leaves faithfulness, while $C_i = 0$; abstaining responses contribute no claims and are excluded by this rule. Writing $\mathcal{I}_F$ for valid records with defined F_i, $\mathcal{I}_C$ for all valid completeness evaluations, $n_F = |\mathcal{I}_F|$, and $n_C = |\mathcal{I}_C|$,

$$F = \frac{1}{n_F} \sum_{i \in \mathcal{I}_F} F_i, \qquad C = \frac{1}{n_C} \sum_{i \in \mathcal{I}_C} C_i. \tag{2}$$

We report the harmonic mean F–C F1 $= 2FC/(F + C)$. Records excluded from F still contribute $C_i = 0$, so abstaining on an answerable record cannot inflate F–C F1. Per-record computation keeps records with more claims or key points from dominating the aggregates. We also report mean claims per response.

Three rates test whether the model respects what the evidence permits: the scope-violation rate $\text{SVR} = N_{\text{P}}^{\text{sv}}/N_{\text{P}}$, where a violation means answering unsupported scope substantively rather than omitting it or acknowledging the gap; the over-abstention rate $\text{OAR} = N_{\text{Y}}^{\text{abs}}/N_{\text{Y}}$, which with SVR bounds scope behavior from both sides; and the clarification rate $\text{CR} = N_{\text{A}}^{\text{clar}}/N_{\text{A}}$, where a response clarifies if it requests missing information, preserves multiple interpretations, or names the ambiguity. On the clean YES/NO subset, with abstention as the positive class, $P_{\text{abs}} = N_{\text{N}}^{\text{abs}}/(N_{\text{N}}^{\text{abs}} + N_{\text{Y}}^{\text{abs}})$ and $R_{\text{abs}} = N_{\text{N}}^{\text{abs}}/N_{\text{N}}$ give F1_{abs} as their harmonic mean, undefined and reported as zero for a model that never abstains. PARTIAL and AMBIGUOUS records stay separate because they require qualification and clarification rather than a binary answer–abstain decision.

C. Judge Configuration and Validation

We instantiate Claude Sonnet 4.6 at temperature 0. The judge sees the query, model response, and reviewed reference response for YES/PARTIAL; only the response for NO; and the query and response for AMBIGUOUS. We validate it using one label-conditioned binary verdict for each whole response. A blinded human reviewer independently evaluates 200 stratified record–response pairs (60 YES, 60 PARTIAL, 40 NO, and 40 AMBIGUOUS) using the same criterion: a YES/PARTIAL answer is unfaithful if it contains an unsupported claim, a NO

TABLE III
GROUNDED ANSWER QUALITY ON 2,998 YES/PARTIAL RECORDS. n_F COUNTS RESPONSES ENTERING FAITHFULNESS; CLAIMS IS THE MEAN ATOMIC-CLAIM COUNT. COMPLETENESS USES ALL VALID EVALUATIONS. BOLD MARKS THE BEST SCORE.

Model	n_F	Claims	Faith. ↑	Compl. ↑	F–C F1 ↑
DeepSeek	2881	6.65	**78.54**	63.79	70.40
Qwen3.5	2956	8.55	67.37	77.74	72.18
Gemma	2970	10.27	69.89	**84.97**	**76.70**
Qwen3	2974	7.37	64.92	60.03	62.38

answer if it supplies substantive content, and an AMBIGUOUS answer if it commits to one interpretation without flagging the ambiguity. A pair counts as agreement when the judge and reviewer assign the same binary verdict; they agree on 189 pairs (94.5%). We report Wilson 95% confidence intervals for binomial behavior rates; AMBIGUOUS results remain descriptive because this class contains only 48 records.

VI. BASELINE EVALUATION ON VIFAITHRAG

A. Baseline Models

We evaluate four instruction-tuned baseline models. They are DeepSeek-V4-Flash, Gemma-4-31B-IT, Qwen3-8B, and Qwen3.5-397B-A17B. Their total parameter counts are 284B, 30.7B, 8.2B, and 397B, respectively; DeepSeek and Qwen3.5 have 13B and 17B active parameters, respectively [15]–[18].

B. Evaluation Setup

Each model receives the original query and the untruncated full text of the same three retrieved pages used to construct the benchmark. The models are not given the answerability label, supporting-document annotations, or reviewed reference response. Generation uses a temperature of 0 and a maximum of 2,048 output tokens. Reasoning modes are disabled, and only the final answer is retained for evaluation.

C. Main Results

1) Grounded Answer Quality: Table III reports grounded answer quality across all 2,998 YES/PARTIAL records. DeepSeek-V4-Flash is the most faithful (78.54%), whereas Gemma-4-31B-IT has the highest completeness (84.97%) and F–C F1 (76.70%). Qwen3.5-397B-A17B is intermediate in completeness and abstention recall but does not interpolate the faithfulness ordering, ranking below Gemma despite its larger total parameter count. Qwen3-8B has the lowest completeness and F–C F1.

The differing n_F values reflect confirmed YES abstentions and, for Gemma, five zero-claim responses. Completeness retains those records. Faithfulness therefore measures supported content without treating refusal statements as substantive claims.

2) Evidence-Boundary Behavior: Table IV evaluates whether models answer, qualify, or abstain at the correct evidence boundary. Supports are 335 PARTIAL records for SVR, 2,663 YES records for OAR, 2,663 YES and 1,071 NO records for abstention, and 48 AMBIGUOUS records for clarification.

The second-pass YES audit identifies 206 unnecessary abstentions across four models: 117 for DeepSeek, 42 for Qwen3.5, 24 for Qwen3, and 23 for Gemma. DeepSeek has the best scope control and abstention recall, but its frequent YES abstentions reduce precision to 86.76%. Gemma minimizes over-abstention and leads precision, while Qwen3.5 achieves the best abstention F1. Qwen3 violates unsupported PARTIAL scope on 59.40% of records and abstains on only 20.35% of NO records. Clarification is also unreliable: the best AMBIGUOUS clarification rate is 52.08%.

3) Response Strategy and Domain Effects: Claim volume helps explain the faithfulness–completeness trade-off. Gemma produces 10.27 atomic claims per answer, followed by Qwen3.5 (8.55), Qwen3 (7.37), and DeepSeek (6.65). Gemma covers more reference points than DeepSeek but has a lower supported-claim fraction. Across legal and open-domain subsets, faithfulness differs by at most 5.13 points per model, whereas abstention F1 differs by as much as 13.12 points. These results support reporting answer quality separately from the prior decision of whether the evidence warrants an answer.

VII. ETHICAL CONSIDERATIONS

Source queries and retrieved documents are publicly accessible; annotations, reviewed reference responses, and model outputs were created for this study. Public availability does not waive source terms, copyright, or privacy obligations. Model responses may be inaccurate or misleading and should not be treated as verified information or professional advice, particularly for legal questions; downstream use requires independent verification and human oversight.

VIII. LIMITATIONS

ViFaithRAG represents one language, one retrieval configuration, and one web snapshot. Changing the search engine, retrieval depth, location, or crawl date may change answerability. The AMBIGUOUS class has only 48 records, and the LLM-assisted construction process may retain model-family bias despite human final decisions. Because the queries and retrieved pages are publicly crawled web content, they may appear in the pretraining corpora of the baseline models. This does not affect the released labels, which record whether the supplied pages support an answer rather than whether an answer is true, but a substantive response on a NO record may reflect parametric recall rather than fabrication, and completeness may be inflated for models with greater exposure to Vietnamese web text.

The automatic judge compares outputs with reviewed reference responses rather than rereading full pages. Its 94.5% whole-answer agreement supports aggregate use, but correct details omitted from a concise reference response may be penalized, and the claim-level decomposition has not been separately validated by humans.

IX. CONCLUSION AND FUTURE WORK

ViFaithRAG evaluates faithful Vietnamese RAG where real queries meet noisy or insufficient web retrieval. Its graded

TABLE IV
EVIDENCE-BOUNDARY BEHAVIOR (%; WILSON 95% INTERVALS FOR BINOMIAL RATES). OAR IS OVER-ABSTENTION ON YES RECORDS; F1 IS SHOWN WITHOUT AN INTERVAL. BOLD MARKS THE BEST POINT ESTIMATE.

	Scope behavior		Abstention (YES/NO)			Ambiguity
Model	SVR ↓	OAR ↓	P ↑	R ↑	F1 ↑	Clar. ↑
DeepSeek	**22.39** (18.25, 27.15)	4.39 (3.68, 5.24)	86.76 (84.37, 88.84)	**71.62** (68.84, 74.23)	78.47	47.92 (34.47, 61.67)
Qwen3.5	35.52 (30.59, 40.78)	1.58 (1.17, 2.12)	94.59 (92.77, 95.98)	68.63 (65.79, 71.34)	**79.55**	**52.08** (38.33, 65.53)
Gemma	32.54 (27.74, 37.73)	**0.86** (0.58, 1.29)	**96.50** (94.80, 97.66)	59.20 (56.23, 62.10)	73.38	41.67 (28.85, 55.72)
Qwen3	59.40 (54.07, 64.53)	0.90 (0.61, 1.34)	90.08 (85.67, 93.24)	20.35 (18.05, 22.87)	33.21	18.75 (10.19, 31.94)

labels and reviewed reference responses separate grounded generation from the prior decision to answer, qualify, clarify, or abstain. The best observed faithfulness and abstention F1 are 78.54% and 79.55%, respectively, but no model leads both; even the most scope-controlled system answers beyond the supported scope on 22.39% of PARTIAL records. The benchmark and code are available at https://github.com/thanhdat00/ViFaithRAG.

Future work will test whether generation- and retrieval-stage interventions, including chain-of-thought prompting, document pre-filtering, and selective chunk retrieval, improve both grounded answer quality and evidence-boundary behavior. We also plan to extend the benchmark beyond web pages to question answering over documents and structured data such as JSON, while expanding ambiguous cases and validating claim-level judging at a larger scale.

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